University of Dhaka
CSE 2106: Microprocessor and Assembly Language Lab
Lab Evaluation 2
[Ques 1 is mandatory] + [Solve another one based on Your roll%3+2]

1. Write an assembly program that takes an array of n numbers and arranges the array in ascending order.
2. Write an assembly code to count the number of 1s in an 8-bit value, which is stored in memory. If it's even, store 0 (even parity), and if it's odd, store 1 (odd parity).
3. You are given a 32-bit value. Write an assembly code to toggle every even bit (bit positions 0, 2, 4, ...) and store the result. Input value will be in the memory address 0x20000000 and store the result in the memory 0x 20000004.
4. You are given a 32-bit unsigned integer stored in memory. Write an assembly program to count the number of consecutive 0 bits starting from the Least Significant Bit (LSB) and store the count value in memory.
   Trailing zeros mean the number of zeros starting from the rightmost bit (bit-0) until the first 1 is encountered.
   Sample Input 1: 0000 0000 0000 0000 0000 0000 0010 1000 Output: 3
   Sample Input 2: 0000 0000 0000 0000 0000 0010 0000 0000 Output: 9